



Oil Pipeline Leak Detection and Localization Simulator Project Documentation



EEE 4416 MINI-PROJECT · ENGINEERING SIMULATION

A physics-informed pipeline monitoring prototype combining hydraulic simulation, virtual sensing, interpretable anomaly detection, fault classification, localization, validation, and an operator dashboard.

PROJECT AT A GLANCE

- 50 km simulated pipeline
- 11 distributed pressure sensors
- 6 operating/fault scenarios
- 3 diagnostic evidence sources
- 4 fault/condition classes
- 1 integrated operator dashboard



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Executive Summary

Oil pipeline monitoring is fundamentally a problem of turning distributed measurements into an engineering decision: **Is the pipeline behaving normally? If not, what happened, where did it happen, and how strong is the evidence?**

This project develops a controlled simulation environment for that workflow. A simplified 1-D pressurized pipeline model generates virtual pressure and flow telemetry under normal and faulted conditions. Configurable measurement noise is introduced to make the diagnostic problem more representative of imperfect sensing.

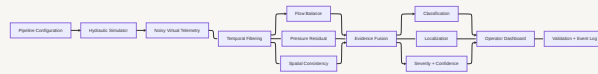
The detector does not depend on a black-box classifier. Instead, it combines three physically interpretable signals, **flow balance, pressure residual, and spatial consistency**, into a transparent evidence score. The result is presented through a PySide6 operator dashboard with alarms, telemetry plots, scenario controls, localization, confidence indicators, ground-truth validation, event logging, and CSV export.

✓ **Core engineering idea:** use multiple independent physical signatures to turn noisy sensor telemetry into an interpretable monitoring decision.

01 · System Overview

What the prototype does

The complete workflow is:



The architecture deliberately separates the **physical model**, **sensor layer**, **diagnostic layer**, **decision layer**, and **operator interface**. This makes the prototype easier to inspect, test, and extend.

Monitoring concept

Layer	Function	Output
Physical model	Simulates pipeline behavior	Pressure + flow state
Sensor model	Adds measurement uncertainty/faults	Virtual telemetry
Filtering	Reduces short-term fluctuations	Smoothed telemetry
Diagnostics	Tests physical consistency	Evidence scores
Fusion	Combines evidence	Anomaly score
Classification	Identifies condition	Normal / Leak / Blockage / Sensor Fault
Localization	Finds affected region	Sensor interval
Validation	Compares with simulator truth	Localization error
Dashboard	Presents results	Operator-facing decision support

02 · Pipeline & Sensor Model

The current implementation models a **50 km pressurized pipeline** with **11 distributed pressure-sensor positions**, together with inlet and outlet flow measurements.

The nominal pressure behavior is represented using a simplified hydraulic relationship:

$$P(x) = P_{in} - RQ^2x$$

where $P(x)$ is pressure at position x , P_{in} is inlet pressure, R is a simplified hydraulic resistance parameter, and Q is flow rate.

Flow consistency is represented by:

$$\Delta Q = Q_{in} - Q_{out}$$

For nominal operation, the inlet/outlet difference should remain small relative to the simulated measurement uncertainty. A persistent imbalance provides evidence of an unexplained loss mechanism.

The hydraulic model is intentionally simplified and steady-state. The parameter values are simulation parameters, not calibrated properties of a specific real-world pipeline.

Reference configuration

Parameter	Value
Pipeline length	50 km
Pressure sensors	11
Reference inlet pressure	70
Reference outlet pressure	20
Reference flow	500
Pressure noise σ	0.2
Flow noise σ	5.0
Simulation step	0.5 s

03 • Fault Scenarios

The simulator provides a controlled test environment with six distinct operating conditions.

Scenario	Simulated condition	Expected diagnostic behavior
Normal Operation	Nominal pipeline + sensor noise	Stable baseline
Small Leak	~5% reference flow loss	Leak / Small
Moderate Leak	~15% reference flow loss	Leak / Moderate
Major Leak	~30% reference flow loss	Leak / Major
Blockage	Severe flow restriction	Blockage
Sensor Fault	Isolated pressure anomaly	Sensor Fault

This scenario structure allows the same diagnostic architecture to be tested against both **physical pipeline anomalies** and **instrumentation anomalies**.

04 · Detection Engine

4.1 Flow-Balance Evidence

The detector evaluates the difference between inlet and outlet flow. A significant unexplained difference is a direct indicator of flow loss within the simplified model.

$$\Delta Q = Q_{in} - Q_{out}$$

Flow evidence receives the highest weight because it provides direct information about a loss mechanism.

4.2 Pressure-Residual Evidence

For each pressure sensor, the detector compares measured pressure with expected pressure:

$$r_i = P_{measured,i} - P_{expected,i}$$

The residual is normalized against an effective noise estimate. This creates a spatial diagnostic signal rather than relying only on total flow.

4.3 Spatial Consistency

Pressure residuals are evaluated as a spatial pattern. Smoothing and spatial-gradient information help identify a transition between less-affected and more-affected regions of the pipeline.

4.4 Temporal Filtering

An exponential moving average reduces short-term measurement fluctuations before diagnostic processing. This improves stability while retaining changes relevant to fault detection.

4.5 Evidence Fusion

The current transparent fused score is:

$$S = 0.45S_{flow} + 0.35S_{pressure} + 0.20S_{spatial}$$

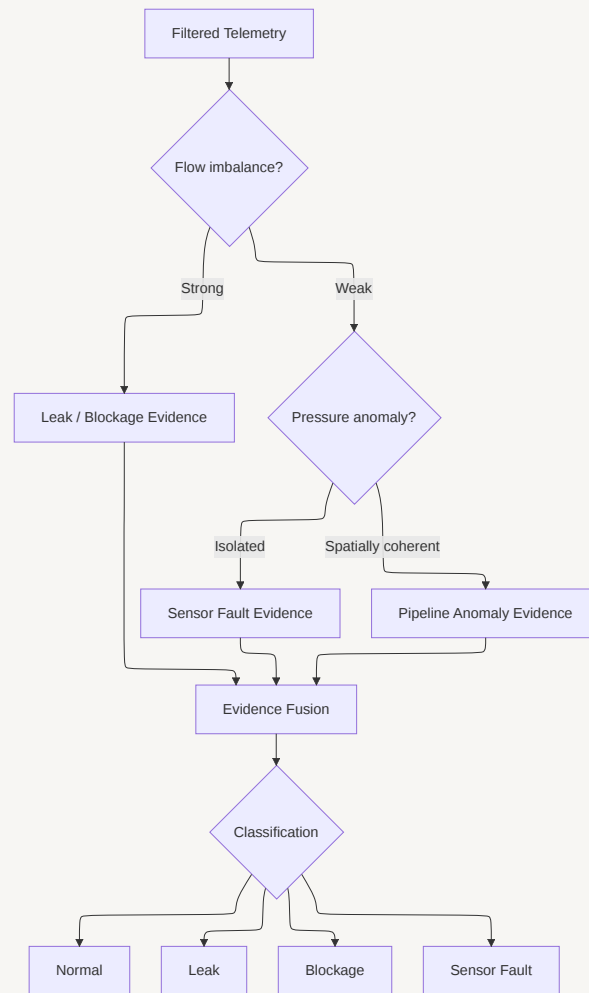
Evidence source	Weight	Engineering role
Flow balance	45%	Detect unexplained flow loss
Pressure residual	35%	Identify hydraulic deviation
Spatial consistency	20%	Support spatial reasoning/localization



The weights are **simulation-oriented design parameters**, not calibrated industrial probabilities. Their purpose is transparency and reproducibility.

05 • Decision Logic

The detector combines evidence to distinguish four principal outcomes.



A leak requires sufficient fused anomaly evidence together with meaningful flow evidence. A blockage is associated with severe flow reduction. An isolated pressure anomaly without strong flow evidence can indicate an instrumentation fault.

06 • Localization

The prototype intentionally avoids claiming more precision than its sensor spacing and simplified model can support.

Instead of reporting an exact leak coordinate, the detector identifies the **most likely affected sensor interval** and reports its midpoint.

The localization error is calculated as:

$$E_{loc} = |x_{estimated} - x_{true}|$$

This distinction is important: the current system demonstrates **segment-level localization**, not field-validated point-location accuracy.

07 · Operator Dashboard

The dashboard is the main operator-facing component of the project. It turns the numerical simulation and detector output into an integrated monitoring interface.

Normal operation



Normal-operation dashboard

Figure 1, Operator dashboard during normal pipeline operation.

The normal state provides the baseline against which abnormal telemetry can be interpreted.

Leak detection



Leak dashboard

Figure 2 , Dashboard response during a simulated leak condition.

The interface brings together pipeline status, telemetry, anomaly information, classification, severity, confidence, and estimated affected location.

Validation view



Validation dashboard

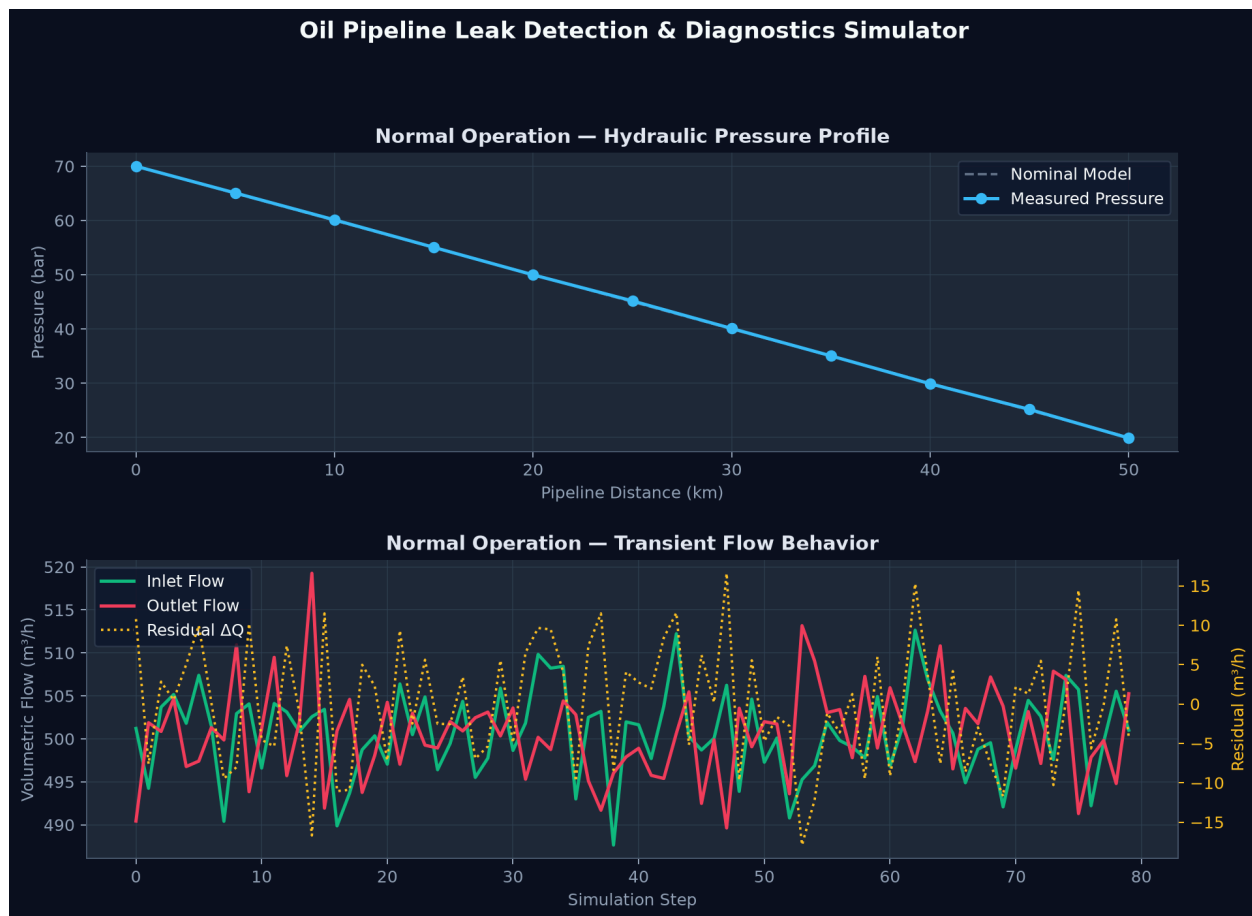
Figure 3 , Dashboard view showing validation and localization information.

The validation view allows the estimated location to be compared against simulator ground truth and the resulting localization error to be inspected.

08 · Simulation Results

The repository contains generated figures for every major scenario. Together they demonstrate that the same diagnostic framework is applied across different operating conditions.

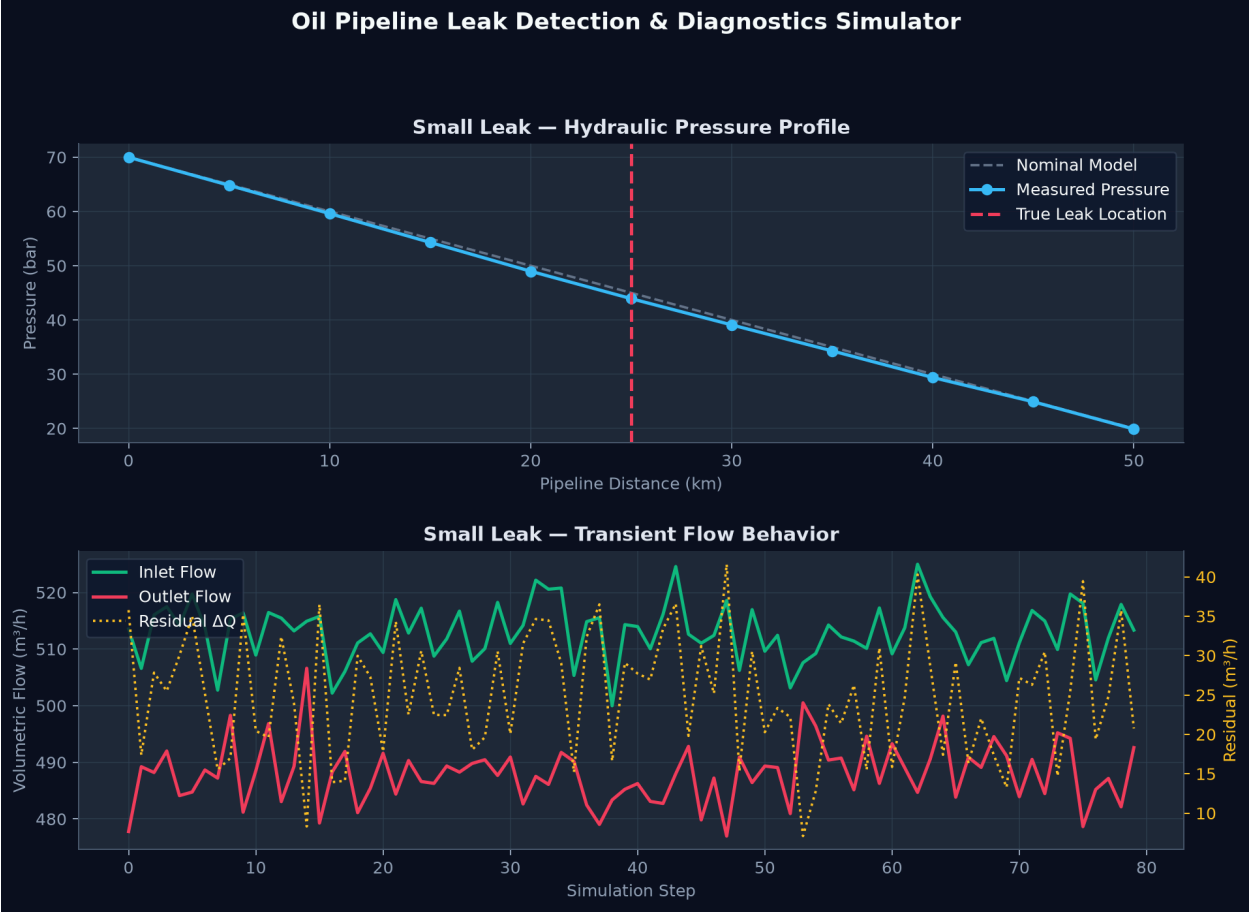
Normal Operation



Normal operation result

Figure 4 , Simulated telemetry and diagnostic behavior under normal operation.

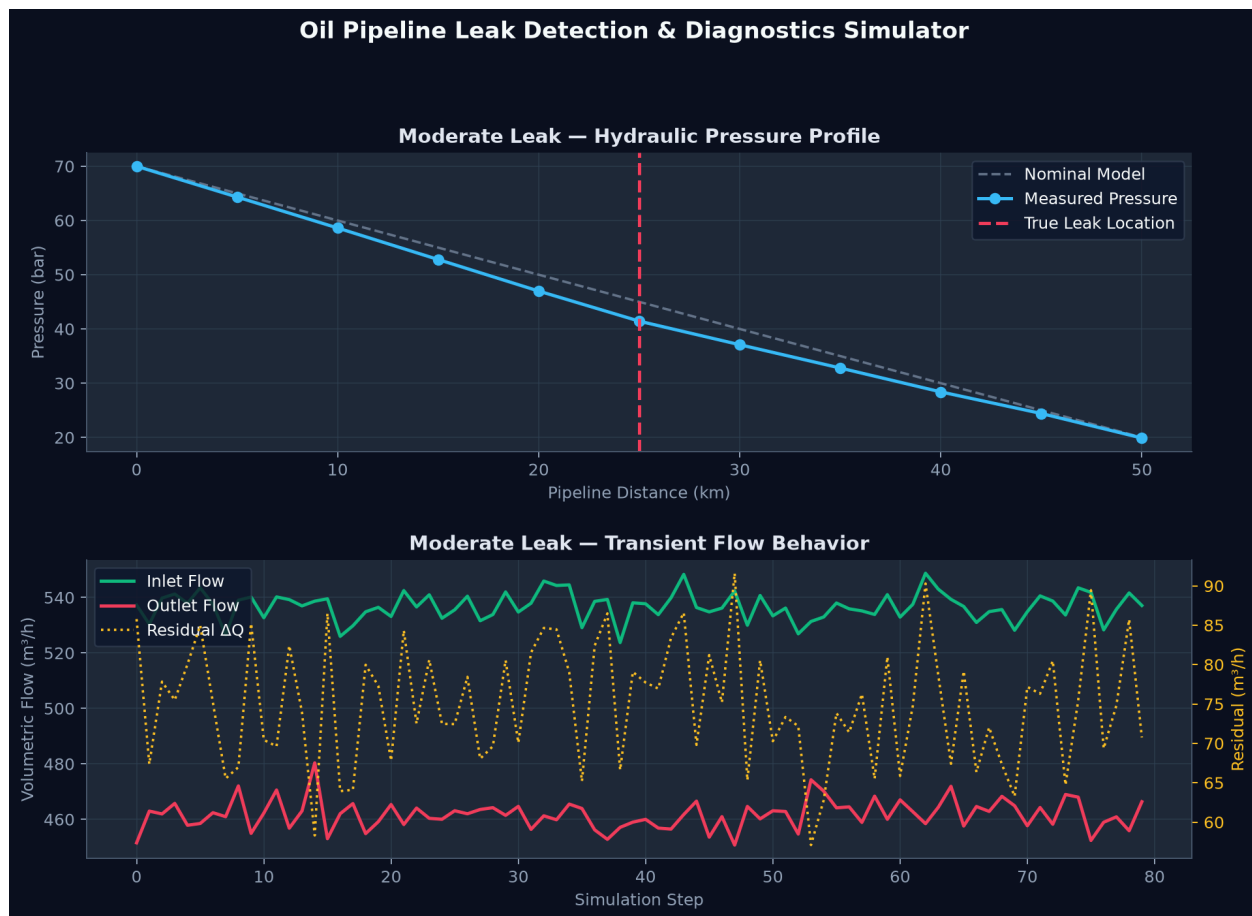
Small Leak



Small leak result

Figure 5 , Simulated small-leak condition.

Moderate Leak



Moderate leak result

Figure 6 , Simulated moderate-leak condition.

Major Leak

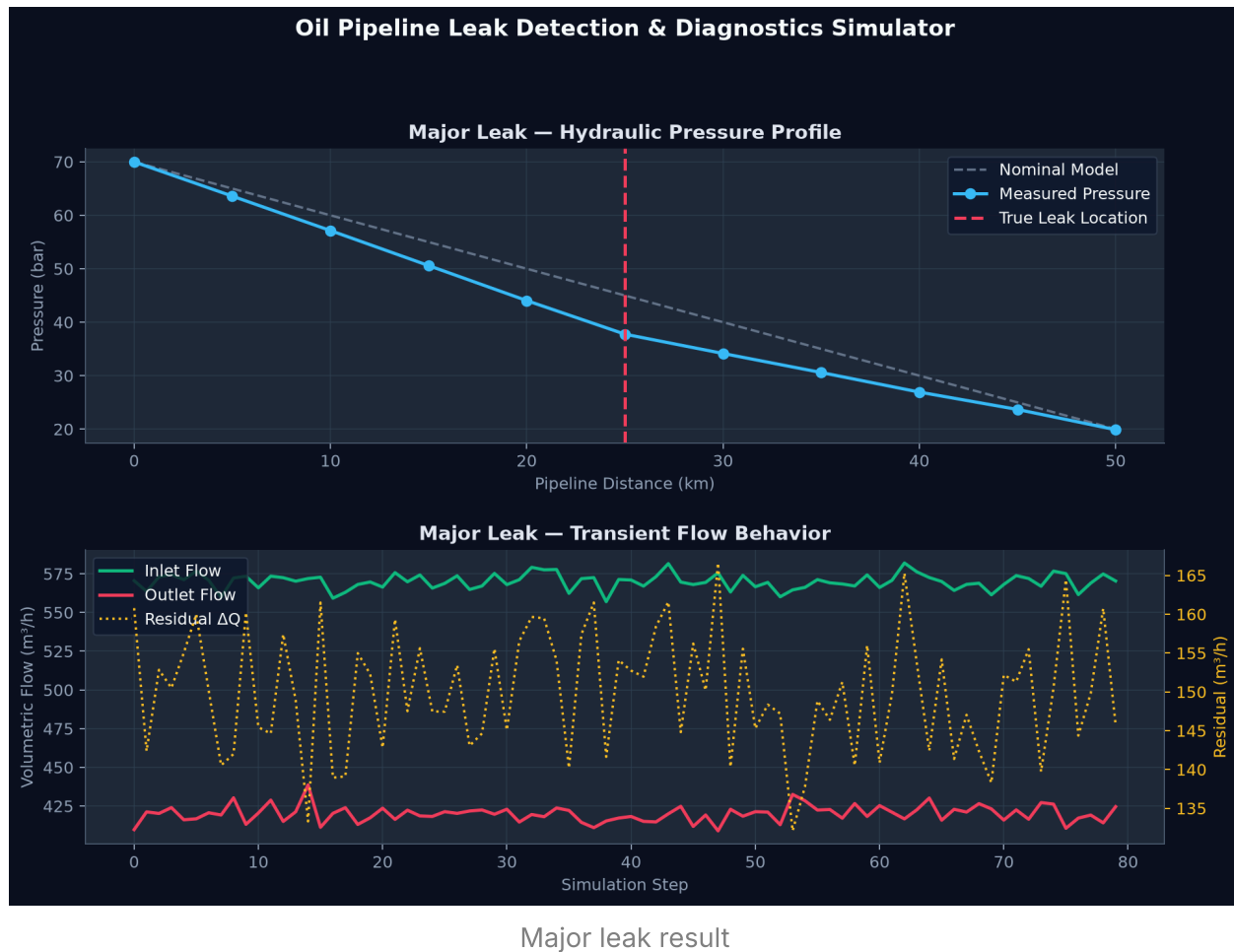


Figure 7, Simulated major-leak condition.

Blockage

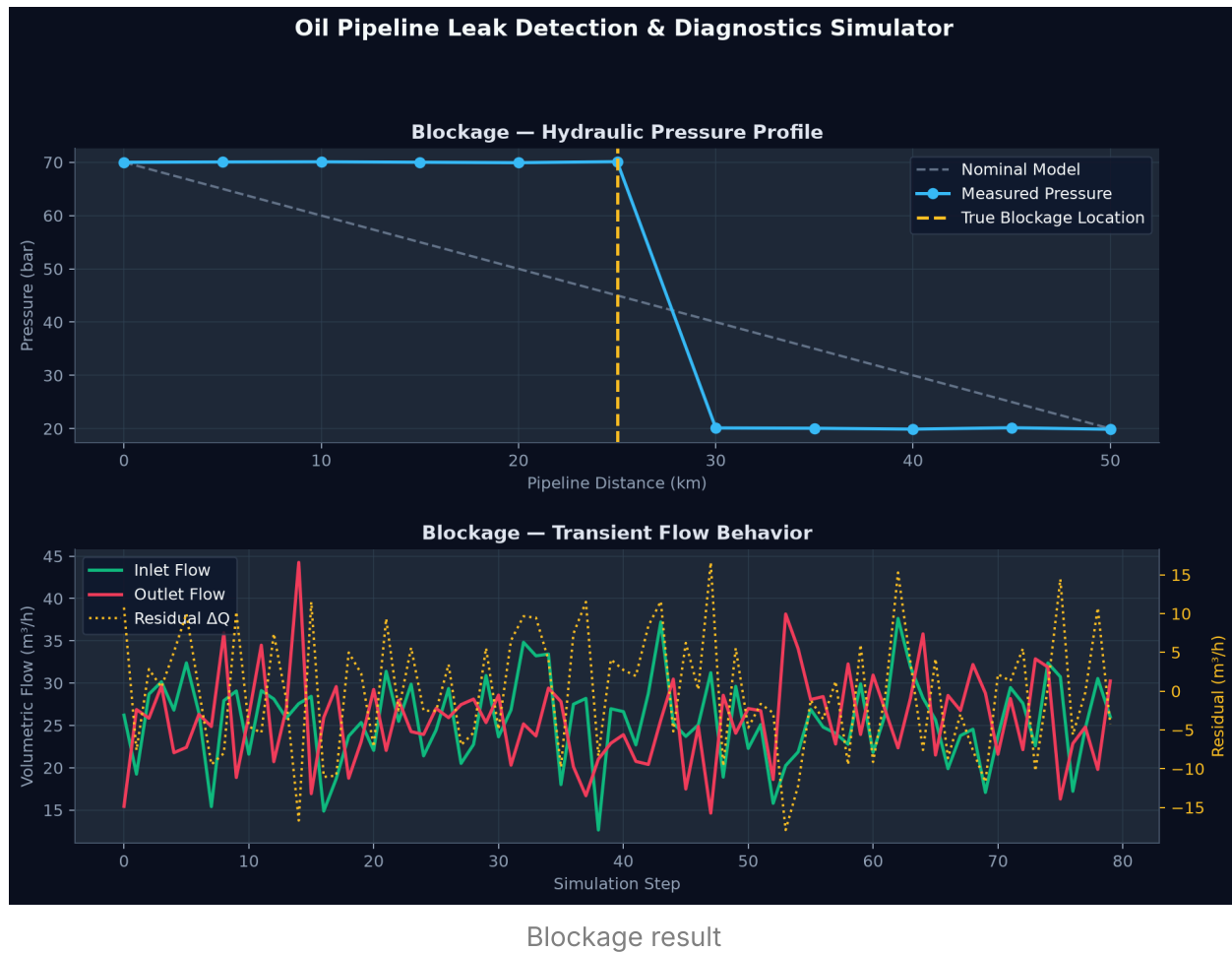


Figure 8 , Simulated blockage condition.

Sensor Fault

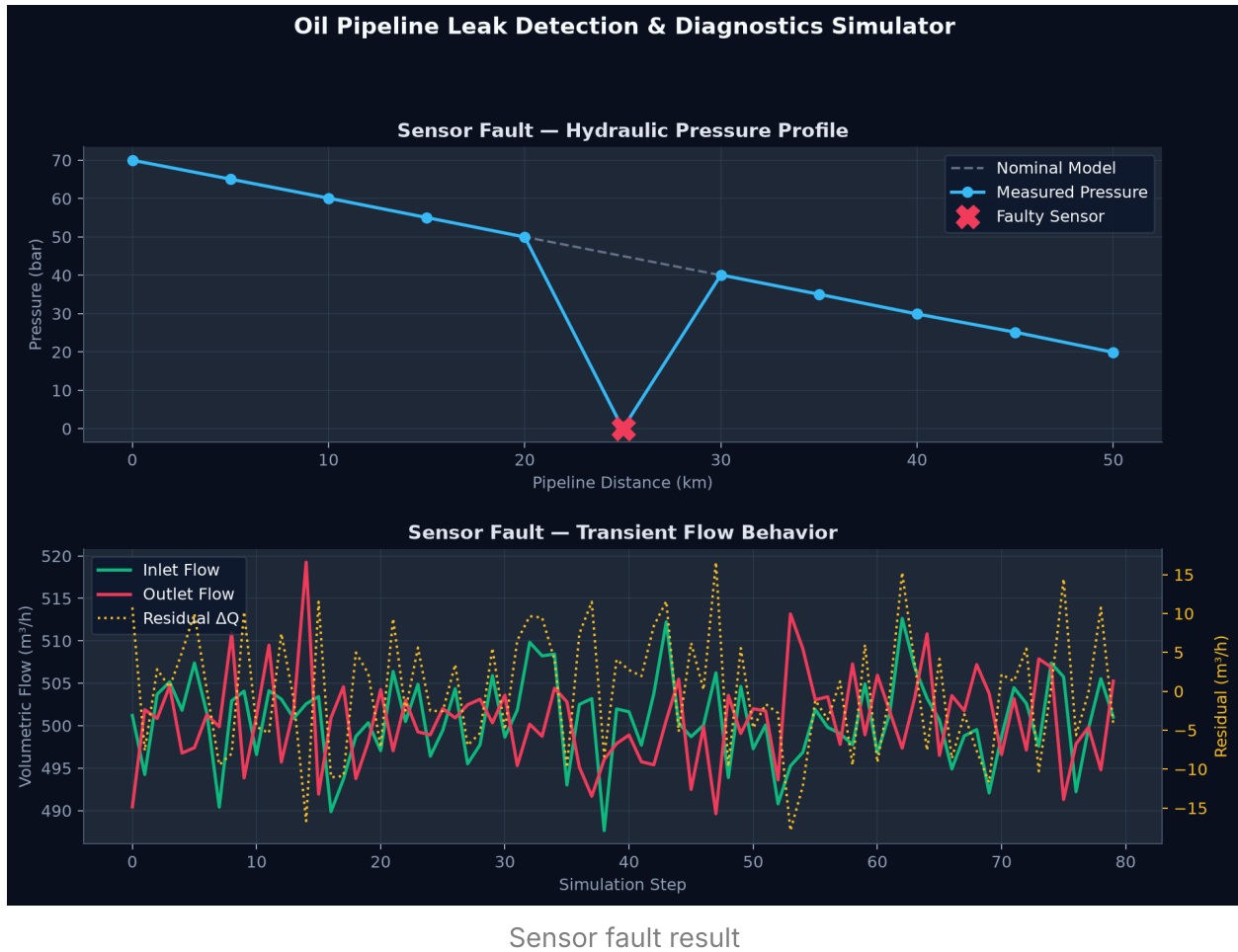


Figure 9 , Simulated isolated sensor-fault condition.

09 · Severity & Confidence

The prototype estimates severity from relative flow loss.

Category	Approximate simulated flow loss
Small	< 10%
Moderate	10–25%
Major	> 25%

The confidence indicator expresses the strength and consistency of the available evidence. It is a **decision-support indicator**, not a calibrated probability of failure.

10 • Validation Framework

The simulator retains injected scenario information and, where applicable, fault location as ground truth. This information is used for evaluation rather than supplied to the detector.

Core evaluation metrics

Metric	Purpose
Detection rate / recall	Measures successful event detection
False-alarm rate	Measures incorrect anomaly alarms
Missed-detection rate	Measures undetected events
Classification accuracy	Measures scenario identification
Localization error	Measures distance from injected location
Localization RMSE	Measures repeated-trial localization performance

Validation workflow

1. Establish a stable normal-operation baseline.
2. Inject each fault scenario separately.
3. Repeat faults at different pipeline locations.
4. Repeat tests at different noise levels.
5. Record classification, severity, confidence, and estimated location.
6. Compare detector output against simulator ground truth.
7. Calculate detection and localization metrics.
8. Export representative telemetry for reproducibility.

11 • Engineering Interpretation

The project demonstrates several important principles used in engineering monitoring systems.

Redundant evidence. A single sensor or signal can be ambiguous. Combining flow, pressure, and spatial information provides complementary evidence.

Physical consistency. The detector asks whether measurements are consistent with the expected hydraulic behavior rather than simply searching for arbitrary patterns.

Explainability. The anomaly score can be decomposed into its flow, pressure, and spatial contributions.

Separation of diagnosis and validation. Ground truth is retained for evaluation without being used as detector input.

Operator-oriented design. The final output communicates status and evidence through an interface rather than leaving the user with raw numerical arrays.

12 · Software Architecture

Component	Responsibility
<code>sim_engine.py</code>	Pipeline state, operating conditions, noise, and fault injection
<code>detector.py</code>	Filtering, diagnostics, evidence fusion, classification, localization
<code>dashboard.py</code>	PySide6 operator interface and visualization
<code>generate_results.py</code>	Automated scenario/result generation
<code>results/figures/</code>	Generated scenario figures
<code>screenshots/</code>	Dashboard evidence
<code>docs/</code>	Technical methodology, validation, limitations, and project documentation

Execution

```
python3 dashboard.py
```

The repository contains the complete implementation, requirements, documentation, generated figures, sample telemetry, and supporting assets.

13 • Limitations

The current prototype is intentionally scoped as an academic engineering simulation.

- The hydraulic model is simplified, steady-state, and one-dimensional.
- It does not implement full transient hydraulic equations.
- Fluid properties and detailed pipeline characteristics are simplified.
- Sensor behavior is represented through simplified noise and fault injection.
- Fault signatures are generated synthetically.
- Detection thresholds and evidence weights are heuristic.
- Localization is limited to the most likely sensor interval.
- Confidence values are not statistically calibrated.
- No industrial benchmark or field dataset has been used.
- The system is not intended for operational safety decisions.

These limitations define the current scope and provide clear directions for further development.

14 • Future Development Roadmap



Potential extensions include transient hydraulic modeling, negative-pressure-wave detection, realistic leak hydraulics, statistical change-point detection, Monte Carlo robustness analysis, detection/false-alarm statistics, localization RMSE, sensor-placement optimization, streaming telemetry, digital-twin integration, machine-learning comparison, and validation against real pipeline datasets.

15 • Conclusion

This project implements an end-to-end pipeline monitoring workflow: **physical simulation** → **virtual sensing** → **filtering** → **engineering diagnostics** → **evidence**

fusion → classification → localization → validation → operator visualization.

Its main contribution is not a claim of industrial deployment readiness. Rather, it demonstrates how a physically informed and interpretable monitoring architecture can be constructed, tested, visualized, and evaluated in a reproducible software environment.

The resulting system provides a strong foundation for extending the work toward higher-fidelity hydraulic simulation, statistical performance evaluation, intelligent monitoring, and eventually real telemetry or digital-twin integration.



Project repository: [Oil Pipeline Leak Detection and Localization Simulator](#)

The GitHub repository contains the executable implementation, figures, screenshots, sample telemetry, and supporting technical documentation.

Appendix · Project Parameters

Parameter	Current implementation
Pipeline length	50 km
Pressure sensor count	11
Small leak	5% reference flow loss
Moderate leak	15% reference flow loss
Major leak	30% reference flow loss
Blockage flow factor	5% theoretical flow
Evidence fusion	45% flow + 35% pressure + 20% spatial
Localization output	Most likely sensor interval midpoint
GUI framework	PySide6
Visualization	Matplotlib
Numerical processing	NumPy / Pandas



Academic scope statement: This documentation describes the behavior and assumptions of the implemented simulation. Results demonstrate software behavior within the simulated conditions and should not be interpreted as field-level performance, industrial certification, or operational safety validation.